Submission Summary

Conference Name	11th International Conference on Computer Vision & Image Processing (CVIP 2026)
Track Name	CVIP2026-Phase-I-Resubmission
Paper ID	812
Paper Title	STAKE - Spatial Temporal Attention Keyframe Extractor using CNNs and CLIP Vision Language Model for Kathakali dance
Abstract	Kathakali is a dance form from the Indian state of Kerala where different hand gestures and facial expressions can be combined with body positions and angles to represent words used in daily life. Translating the words from videos of Kathakali dance postures requires keyframes consisting of still postures, which in turn requires means of keyframe identification. In this work, we propose a novel approach for keyframe identification and extraction, called STAKE (Spatial Temporal Attention Keyframe Extractor), created using CNN models, attention and LSTMs. We collect videos for twenty words used in Kathakali and manually identify the keyframes from these videos, which serve as Ground Truth. We then use different CNN models such as ResNet50, DenseNet121, EfficientNetB0 and MobileNetV2 to extract features, combine them with the features identified by the CLIP Vision Language model, and extract the temporal dependencies between image frames using Bi-LSTM. The detected keyframes, on comparison with the Ground Truth, gives 77% accuracy, and comparable precision@9, recall@9, Macro F1-score, Weighted F1-score, Mean Absolute Temporal Error (MATE) as well as Dynamic Time Warping (DTW). We further conducted ablation studies as well as compared the performance of our best performing model with the existing baseline models. Further, we also performed Wilcoxon's signed rank test to check if the distribution of the scores for each video differ significantly between the different models. Our future work will concentrate on using the classifying the extracted keyframes to translate the videos of Kathakali dance to natural language words, for use in digital archival as well as heritage preservation.
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Primary Subject Area	Image/ Video Processing

Submission Files	STAKE - Spatial Temporal Attention Keyframe Extractor using CNNs and CLIP Vision Language Model for Kathakali dance.pdf (11.7 Mb, 2026-07-22 15:51:52 GMT+5:30) Response to Reviewers.pdf (465 Kb, 2026-07-22 15:51:57 GMT+5:30)
Submission Questions Response	**1. Phase-I Paper ID** *Enter the Phase-I Paper ID* *Note: This field is only for papers that received a "Revision and Resubmit" decision during the CVIP 2026 Phase-I submission. All other papers should be submitted under the "CVIP 2026 – Phase-II" track.* 339